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United States Patent	12398408
Kind Code	B2
Date of Patent	August 26, 2025
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Recombinant *Escherichia coli* for producing glutarate, construction method and use thereof

Abstract

The present invention provides recombinant *Escherichia coli* for producing glutarate, a construction method and use thereof. A double-plasmid recombinant bacterium is constructed through molecular biological means for co-expressing an aldehyde synthase (AAS) gene, an amine oxidase Mao (gene) and an aldehyde dehydrogenase (Glox) gene. The constructed expression plasmids are introduced into the *Escherichia coli* to reconstruct to obtain recombinant cells. A recombination strain for efficiently producing glutarate is obtained through amicillin resistance and kanamycin resistance combined plate screening. Efficient production of the glutarate is achieved by optimizing concentration of a substrate, cell concentration and a transformation temperature. L-lysine with a concentration of 30 g/L may be transformed into 19.65 g of glutarate through reactions for 30 h under transformation conditions that the cell concentration is 30 g/L, the pH value is 8 and 6 mM of NAD.sup.+ is additionally added, wherein a transformation rate may be 65.3%.

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Appl. No.:	**17/640807**
Filed (or PCT Filed):	**December 03, 2021**
PCT No.:	**PCT/CN2021/135262**
PCT Pub. No.:	**WO2023/092632**
PCT Pub. Date:	June 01, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240043884 A1	Feb. 08, 2024

Foreign Application Priority Data

CN	202111433195.5	Nov. 29, 2021

Publication Classification

Int. Cl.: **C12P7/44** (20060101); **C12N1/21** (20060101); **C12N9/02** (20060101); **C12N9/06** (20060101); **C12N9/88** (20060101); **C12N15/70** (20060101)

U.S. Cl.:

CPC **C12P7/44** (20130101); **C12N9/0008** (20130101); **C12N9/0022** (20130101); **C12N9/88** (20130101); **C12N15/70** (20130101); **C12Y102/01005** (20130101); **C12Y104/03004** (20130101); **C12Y401/01028** (20130101); C12N2800/101 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

(1) This application is the National Stage Application of PCT/CN2021/135262, filed on Dec. 3, 2021, which claims priority to Chinese Patent Application No. 202111433195.5, filed on Nov. 29, 2021, which is incorporated by reference for all purposes as if fully set forth herein.
FIELD OF THE INVENTION
(2) The present invention relates to the technical field of metabolic engineering, and more particularly to a recombinant *Escherichia coli* for efficiently producing glutarate, and a construction method and use thereof.
DESCRIPTION OF THE RELATED ART
(3) Glutarate, commonly known as colloidal acid, is an aliphatic dicarboxylic acid with a molecular formula of C.sub.5H.sub.8O.sub.4 and a molecular weight of 132.11. It is a colorless needle-like crystalline solid at the room temperature, and is freely soluble in water, ethanol, ether, etc., with solubility in water of 430 g/L. Among all dicarboxylic acids, the glutarate is more suitable for production of polyesters and polyamides of nylon-4,5 and nylon-5,5 and the like due to the lowest melting point of 95-98° C. In addition, the glutarate is also a precursor of 1,5-pentanediol which is a common plasticizer of a soldering flux, an activator and an important medical intermediate. In short, the glutarate, as an important C5 platform compound, has an important application value and a development potential in the fields of medicine, chemical synthesis, and the like.
(4) At present, the main preparation method of the glutarate is a chemical synthesis method, in which industrial production mainly refers to recycling from a mixture of oxidized cyclohexanone and cyclohexanol under the catalysis of nitric acid. A small dose of glutamate also can be prepared at the laboratory level, for example, the glutarate is prepared through a series of chemical reactions by taking γ-butyrolactone, dihydropyran, glutaronitrile, cyclohexanone and the like as a substrate. The conventional chemical method for synthesizing the glutarate has the disadvantages of high cost, severe pollution, high requirements on operation conditions, and the like. As a result, it is of profound significance in environmental protection, efficient production and a good production prospect of the glutarate in the future.
(5) In recent years, domestic and foreign researchers have explored and researched the production of the glutarate by microorganisms from two aspects: biochemical engineering and metabolic engineering. So far, there are four biosynthetic pathways of the glutarate reported in the literature, namely: a glutaconate reduction pathway, a carbon chain extension and decarboxylation pathway, a reverse adipate-degradation pathway, and a lysine degradation pathway (including a degradation pathway with pentane diamine as an intermediate and a degradation pathway with 5-aminovaleric acid as an intermediate). All of the pathways adopt excessive enzymes and are too complex. Therefore, how to construct a brand-new and shortest pathway to improve the yield of the glutarate has become an urgent problem to be solved, and it is also one of the research hotspots worldwide.

SUMMARY OF THE INVENTION

(6) To solve the technical problems, the present invention provides a brand-new pathway for producing glutarate, and the effectiveness of the pathway is verified by constructing a single enzyme expression strain. The present invention further provides a recombinant *Escherichia coli* engineering bacterium for efficiently producing glutarate. By constructing a double-plasmid expression system, a gene AAS for coding aromatic aldehyde synthase, a gene Mao for coding amine oxidase and a gene Glox for coding aldehyde dehydrogenase are co-expressed in a host of the *Escherichia coli* to obtain a recombinant bacterium. By taking the constructed recombinant bacterium as a catalyst and L-lysine as a substrate to achieve catalyzed synthesis of the glutarate, the reaction conditions are optimized.

(7) A first object of the present invention is to provide recombinant *Escherichia coli* for producing glutarate. The recombinant *Escherichia coli* is obtained by expressing the aromatic aldehyde synthase (AAS) gene, the amine oxidase (Mao) gene and the aldehyde dehydrogenase (Glox) gene in the host of *Escherichia coli*.

(8) In the present invention, a specifically constructed metabolic reaction path includes transforming the L-lysine into 5-amino valeraldehyde through the aromatic aldehyde synthase (AAS), transforming the 5-amino valeraldehyde into glutaraldehyde through the amine oxidase (Mao) and transforming the glutaraldehyde into the glutarate through the aldehyde dehydrogenase (Glox).

(9) Preferably, the aromatic aldehyde synthase (AAS) gene, the amine oxidase (Mao) gene and the aldehyde dehydrogenase (Glox) gene are expressed through the double-plasmid expression system.

(10) Preferably, the aromatic aldehyde synthase gene and the amine oxidase gene are expressed through the same plasmid.

(11) Preferably, the double-plasmid expression system is a combination of two of a pETM6R1 plasmid, pET28a plasmid, a PRSF plasmid, a pCOR plasmid, a pCDF plasmid and a PACYC plasmid.

(12) In the present invention, in a specific embodiment: the pETM6R1 plasmid is used for expressing the aromatic aldehyde synthase (AAS) gene and the amine oxidase (Mao) gene, and the pET28a plasmid is used for expressing the aldehyde dehydrogenase (Glox) gene.

(13) Preferably, an amino acid sequence of the aromatic aldehyde synthase is as shown in SEQ ID NO:1, an amino acid sequence of the amine oxidase (Mao) is as shown in SEQ ID NO:2, and an amino acid sequence of the aldehyde dehydrogenase (Glox) is as shown in SEQ ID NO:3.

(14) Preferably, the host of the *Escherichia coli* is *E. coli* BL21 (DE3), *E. coli* JM109 (DE3) or *E. coli* MG1655 (DE3).

(15) A second object of the present invention is to provide a construction method of the recombinant *Escherichia coli*, including the following steps: S1, ligating the aromatic aldehyde synthase (AAS) gene and the amine oxidase (Mao) gene to a first vector to obtain a first recombinant vector; S2, ligating the aldehyde dehydrogenase (Glox) gene to a second vector to obtain a second recombinant vector; and S3, introducing the first recombinant vector and the second recombinant vector into the host of the *Escherichia coli* to screen a recombinant bacterium, which is capable of expressing the aromatic aldehyde synthase (AAS) gene, the amine oxidase (Mao) gene and aldehyde dehydrogenase (Glox) gene.

(16) In the present invention, in a specific embodiment: a coding gene of the aromatic aldehyde synthase (AAS) with the amino acid sequence as shown in SEQ ID NO:1 and a coding gene of the amine oxidase (Mao) with the amino acid sequence as shown in SEQ ID NO:2 are ligated to a pETM6R1 vector through enzyme digestion after being amplified, to finally construct a plasmid pETM6R1-AAS-Mao. The aldehyde dehydrogenase (Glox) gene with the amino acid sequence shown in SEQ ID NO:3 is ligated to the pET28a vector through the enzyme digestion after being amplified, to finally construct a plasmid pET28a-Glox. The two plasmids are simultaneously introduced into the *E. coli* MG1655 (DE3) to express.

(17) A third object of the present invention is to provide use of the recombinant *Escherichia coli* in producing the glutarate.
(18) Preferably, a whole-cell of the recombinant *Escherichia coli* is taken as a catalyst for transforming L-lysine to produce the glutarate.
(19) Preferably, in a transformation system, the additive amount of the L-lysine is 20-50 g/L.
(20) Preferably, the transformation system further includes 4-8 mM of NAD.sup.+.
(21) Preferably, transformation conditions are as follows: a pH value of 7.5-8.5, a transformation temperature of 28-32° C., and a rotation speed of 150-250 rpm.
(22) Preferably, the whole-cell of the recombinant *Escherichia coli* is obtained by performing induced fermentation on the recombinant *Escherichia coli*, and specifically, a single colony of the recombinant *Escherichia coli* is inoculated in a seed culture medium with ampicillin and kanamycin resistance to culture overnight under 35-38° C. and 180-220 rpm to obtain a seed solution. The seed solution is transferred into a fermentation culture medium with ampicillin and kanamycin resistance in inoculation amount of 1-5% (v/v) to culture under 35-38° C. and 180-220 rpm until OD600 is 0.6-0.8. IPTG with a final concentration of 0.5-1.5 mM is added to induce for 10-15 h under 22-28° C. and 180-220 rpm to obtain a fermentation liquor. The fermentation liquor is centrifuged to collect a bacterial cell, namely the whole-cell of the recombinant *Escherichia coli*.
(23) Preferably, the seed culture medium includes 8-12 g/L of tryptone, 4-6 g/L of yeast powder, 8-12 g/L of sodium chloride, and the pH value is 7.0-7.2.
(24) Preferably, the fermentation culture medium includes 10-15 g/L of tryptone, 20-25 g/L of yeast powder, 3-5 g/L of glycerol, 12-13 g/L of dipotassium phosphate and 2-3 g/L of monopotassium phosphate.
(25) As compared with the prior art, the invention has the following beneficial effects:
(26) A double-plasmid recombinant bacterium is constructed through molecular biological means for co-expressing the aldehyde synthase (AAS) gene, the amine oxidase Mao (gene) and the aldehyde dehydrogenase (Glox) gene. The constructed expression plasmids are introduced into the *Escherichia coli* to reconstruct to obtain a recombinant cell. A recombination strain for efficiently producing the glutarate is obtained through amicillin resistance and kanamycin resistance combined plate screening. Efficient production of the glutarate is achieved by optimizing concentration of a substrate, cell concentration and the transformation temperature. L-lysine with a concentration of 30 g/L may be transformed into 19.65 g of the glutarate through reactions for 30 h under the transformation conditions that the cell concentration is 30 g/L, a pH value is 8 and 6 mM of NAD.sup.+ is additionally added, where a transformation rate may be 65.3%.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows a glutarate biosynthetic pathway.
(2) FIG. **2** shows the effect of substrate concentration on yield of glutarate.
(3) FIG. **3** shows the effect of NAD.sup.+ on yield of glutarate.
(4) FIG. **4** shows the transformation rate for transforming whole-cells of different bacterial strains.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(5) The present invention will be further described below in conjunction with specific embodiments, so that those skilled in the art may better understand and implement the present invention, but the embodiments described here are not intended to limit the present invention.
(6) Feasibility verification of metabolic pathway of the present invention: a constructed system, namely constructed recombinant *Escherichia coli*, can transform the L-lysine into glutarate with addition of a cofactor NAD.sup.+. The specific reaction process is as shown in FIG. **1**.
(7) In the present invention, the pETM6R1 plasmid, the pET28a plasmid, the RSF plasmid, the

pCOR plasmid, the pCDF plasmid and the PACYC plasmid are commercially available.
(8) In the present invention, *E. coli* BL21(DE3), *E. coli* JM109 (DE3) and *E. coli* MG1655 (DE3) are commercially available.
(9) In the present invention, the seed culture medium includes 10 g/L of tryptone, 5 g/L of yeast powder, 10 g/L of sodium chloride, and the pH value is 7.0-7.2.
(10) The fermentation culture medium includes 12 g/L of tryptone, 24 g/L of yeast powder, 4 g/L of glycerol, 12.53 g/L of dipotassium phosphate and 2.31 g/L of monopotassium phosphate.
(11) In the present invention, the detection method of the glutarate utilizes a HPLC system, an organic acid analytical column (Amine HPX-87H column, 300 mm 7.8 mm) and an ultraviolet detector (210 nm), and 5 mm H.sub.2SO.sub.4 as a mobile phase. The flow rate of the mobile phase is 0.6 mL/min, and the temperature is kept at 60° C. during operation.
Embodiment 1: Construction of Recombinant Plasmid pETM6R1-AAS-Mao
(12) A coding gene of aromatic aldehyde synthase (AAS) with an amino acid sequence as shown in SEQ ID NO:1 and a coding gene of amine oxidase (Mao) with an amino acid sequence as shown in SEQ ID NO:2 were amplified to obtain DNA fragments containing the coding gene of the aromatic aldehyde synthase (AAS) and DNA fragments containing the coding gene of the amine oxidase (Mao). The pETM6R1 plasmid was enzyme digested overnight by the restriction endonuclease XhoI to obtain a linearized vector and a sticky end is exposed. The purified DNA fragments containing the coding gene of the aromatic aldehyde synthase (AAS) and the purified DNA fragments containing the coding gene of the amine oxidase (Mao) were ligated to a pETM6R1 vector through enzyme digestion to reconstruct a recombinant plasmid pETM6R1-AAS-Mao. The reconstructed recombinant plasmid pETM6R1-AAS-Mao was transformed into *E. coli* JM109 competent cells through a chemical transformation method, and the competent cells were cultured for 12 h in an LB flat plate containing ampicillin. PCR verification was performed on a bacterial colony grown on the flat plate. A positive transformant was selected and inoculated into an LB culture medium, and a plasmid was extracted after culturing for 12 h at 37° C. A recombinant plasmid pETM6R1-AAS-Mao was constructed through sequencing verification.
Embodiment 2: Construction of the Recombinant Plasmid pET28a-Glox
(13) A coding gene of aldehyde dehydrogenase (Glox) with an amino acid sequence as shown in SEQ ID NO:3 was amplified to obtain DNA fragments containing the coding gene of the aldehyde dehydrogenase (Glox); the restriction endonucleases BamHI and XhoI were selected for performing double-enzyme digestion on a vector pET28a for 3 h at 37° C. to obtain a linearized vector and a sticky end was exposed. The recycled Glox coding gene was ligated with the plasmid pET28a for 10 h at 16° C. through T4 ligase to reconstruct a recombinant plasmid pET28a-Glox. The constructed recombinant plasmid pET28a-Glox was transformed into *E. coli* JM109 competent cells through a chemical transformation method, and the competent cells were cultured for 12 h in an LB flat plate containing kanamycin. PCR verification was performed on a bacterial colony grown on the flat plate. A positive transformant was selected and inoculated into an LB culture medium, and a plasmid was extracted after culturing for 12 h at 37° C. A recombinant plasmid pET28a-Glox was constructed through sequencing verification.
Embodiment 3: Construction and Expression of Double-Plasmid Recombinant *Escherichia coli*
(14) The recombinant plasmid pETM6R1-AAS-Mao constructed in Embodiment 1 and the recombinant plasmid pET28a-Glox constructed in Embodiment 2 were simultaneously transformed into *E. coli* MG1655 (DE3) competent cells through a chemical transformation method. The competent cells were cultured for 12 h in an LB flat plate with ampicillin resistance and kanamycin resistance. A single colony grown on the flat plate was inoculated into a seed culture medium with ampicillin resistance and kanamycin resistance to culture overnight under 37° C. and 200 rpm. The single colony was transferred into a 100 mL fermentation culture medium with ampicillin resistance and kanamycin resistance at inoculation amount of 2% (v/v), and was culture under 37° C. and 200 rpm until OD600 was 0.6-0.8. IPTG with a final concentration of 1 mM was added for

inducing for 12 h under 25° C. and 200 rpm, and the bacterial cells were collected by centrifuging for whole-cell transformation.

Embodiment 4: Verification of Whole-Cell Transformation of Double-Plasmid Recombinant *Escherichia coli*

(15) A PBS buffer solution was taken as a medium for whole-cell transformation, and transformation conditions were as follows:

(16) Wet cells of the recombinant *Escherichia coli* obtained in Embodiment 3 was centrifuged and used as a catalyst for the whole-cell transformation. Reaction liquid was obtained by rotating for 30 h at 30° C. when the concentration of L-lysine was 30 g/L, the pH value was 8, a rotation speed of a shaker was 220 rpm, a concentration of the cell was 30 g/L. After being centrifuged for 5 min at 12000 rpm, the reaction liquid was filtered by a 0.22 μm filter membrane, was diluted by 10 times and was detected by HPLC. To further verify reliability of the result, the glutarate in the reaction liquid was detected through LC-MS.

Embodiment 5: Optimization of Substrate Concentration for Whole-Cell Transformation of Recombinant *Escherichia coli*

(17) A bacterial strain with a concentration of 30 g/L was added into a 10 mL reaction system, where concentrations of L-lysine was respectively 20 g/L, 30 g/L, 40 g/L and 50 g/L, the pH value was 8, a rotation speed of a shaker was 220 rpm, and transformation lasted for 30 h at 30° C., and the result was as shown in FIG. **2**. When the concentration of L-lysine was 30 g/L, the transformation rate was the highest, up to 60.53%, and the yield was up to 18.59 g.

Embodiment 6: Optimization of Cofactor for Whole-Cell Transformation of Recombinant *Escherichia coli*

(18) A bacterial strain with a concentration of 30 g/L was added into a 10 mL reaction system, where 6 mM of NAD+ was additionally added, the pH value was 8, a rotation speed of a shaker was 220 rpm, transformation lasted for 30 h at 30° C., and the result was as shown in FIG. **3**. When NAD.sup.+ was additionally added, the transformation rate was increased to 65.3%, and the yield was 19.65 g.

Embodiment 7: Construction and Expression of Recombinant Expression Vectors pETM6R1-AAS-MAO, PRSF-AAS-MAO and pCOR-AAS-MAO

(19) On the basis of successfully constructed expression vectors, pathway enzyme was assembled through the ePathBrick technology. The steps were introduced by taking the construction of the recombinant expression vector pETM6R1-AAS-MAO as an example. The restriction endonuclease XhoI was used to obtain a linearized vector and a sticky end was exposed; then, a target gene with the sticky end was then obtained; and finally, the target gene was ligated overnight at 16° C. through T4 ligase, the product was transformed into JM109 competent cells, and a single colony was selected for PCR verification; and the recombinant expression vector pETM6R1-AAS-MAO was successfully constructed if the stripe size was correct, and construction methods of the other two recombinant expression vectors were kept consistent. In such a manner, 3 recombinant expression vectors were obtained, which were respectively pETM6R1-AAS-MAO, PRSF-AAS-MAO and pCOR-AAS-MAO.

Embodiment 8: Construction and Expression of Recombinant Expression Vectors pET28a-GLOX, pCDF-GLOX and PACYC-GLOX

(20) On the basis of successfully constructed expression vectors, pathway enzyme was assembled through the ePathBrick technology. The steps were introduced by taking the construction of the recombinant expression vector pET28a-GLOX as an example. The restriction endonucleases BamHI and XhoI were used to perform double enzyme-digestion on a vector pET28a to obtain a linearized vector and a sticky end was exposed; then, a target gene with the sticky end was then obtained; and finally, the target gene was ligated overnight at 16° C. through T4 ligase, the product was transformed into JM109 competent cells, and a single colony was selected for PCR verification; and the recombinant expression vector pET28a-GLOX was successfully constructed if

the stripe size was correct, and construction methods of the other two recombinant expression vectors were kept consistent. In such a manner, 3 recombinant expression vectors were obtained, which were respectively pET28a-GLOX, pCDF-GLOX and PACYC-GLOX.

Embodiment 9: Construction and Expression of Recombinant Host

(21) According to a scheme of simultaneously expressing an aromatic aldehyde synthase (AAS) gene, an amine oxidase (Mao) gene and an aldehyde dehydrogenase (Glox) gene, the plasmids constructed in Embodiment 8 and Embodiment 9 were respectively transformed into competent cells through a chemical transformation method. The competent cells were respectively *E. coli* BL21 (DE3), *E. coli* JM109 (DE3) and *E. coli* MG1655 (DE3), and a specific combination way of the recombinant expression vectors and a host bacterium is shown in table 1. The constructed recombinant strains were respectively cultured for 12 h in an LB flat plate with resistance, and a single colony grown on the flat plate was inoculated into a seed culture medium with resistance to culture overnight under 37° C. and 200 rpm. The single colony was transferred into a 100 mL fermentation culture medium with ampicillin resistance and kanamycin resistance at an inoculation amount of 2% (v/v), and was cultured under 37° C. and 200 rpm until OD.sub.600 was 0.6-0.8. IPTG with a final concentration of 1 mM was added for inducing for 12 h under 25° C. and 200 rpm, and cells were collected for whole-cell transformation.

(22) TABLE-US-00001 TABLE 1 No. Host Double vectors 1 *E. coli* BL21 pETM6R1-AAS-MAO&pET28a-GLOX 2 (DE3) PRSF-AAS-MAO&pET28a-GLOX 3 pCOR-AAS-MAO&pET28a-GLOX 4 pETM6R1-AAS-MAO&pCDF-GLOX 5 PRSF-AAS-MAO&pCDF-GLOX 6 pCOR-AAS-MAO&pCDF-GLOX 7 pETM6R1-AAS-MAO&PACYC-GLOX 8 PRSF-AAS-MAO&PACYC-GLOX 9 pCOR-AAS-MAO&PACYC-GLOX 10 *E. coli* JM109 pETM6R1-AAS-MAO&pET28a-GLOX 11 (DE3) PRSF-AAS-MAO&pET28a-GLOX 12 pCOR-AAS-MAO&pET28a-GLOX 13 pETM6R1-AAS-MAO&pCDF-GLOX 14 PRSF-AAS-MAO&pCDF-GLOX 15 pCOR-AAS-MAO&pCDF-GLOX 16 pETM6R1-AAS-MAO&PACYC-GLOX 17 PRSF-AAS-MAO&PACYC-GLOX 18 pCOR-AAS-MAO&PACYC-GLOX 19 *E. coli* pETM6R1-AAS-MAO&pET28a-GLOX 20 MG1655 PRSF-AAS-MAO&pET28a-GLOX 21 (DE3) pCOR-AAS-MAO&pET28a-GLOX 22 pETM6R1-AAS-MAO&pCDF-GLOX 23 PRSF-AAS-MAO&pCDF-GLOX 24 pCOR-AAS-MAO&pCDF-GLOX 25 pETM6R1-AAS-MAO&PACYC-GLOX 26 PRSF-AAS-MAO&PACYC-GLOX 27 pCOR-AAS-MAO&PACYC-GLOX

Embodiment 10

(23) A PBS buffer solution was taken as a medium for whole-cell transformation, and transformation conditions were as follows:

(24) Wet cells of 27 recombinant bacteria obtained in Embodiment 9 were respectively selected, centrifuged and used as a catalyst for the whole-cell transformation. When the concentration of the L-lysine was 30 g/L, the cell concentration was 30 g/L, and 6 mM of NAD+ was additionally added, the pH value was 8, the rotation speed of a shaker was 220 rpm, and rotation lasts for 30 h at 30° C. After being centrifuged for 5 min at 12000 rpm, the reaction liquid was filtered by a 0.22 μm filter membrane, was diluted by 10 times and was detected by HPLC, where the transformation rate was as shown in FIG. **4**. The above recombinant bacterial strains were used as a catalyst for whole-cell transformation, and L-lysine was transformed into glutarate. For the transformation rate, there was certain difference among the bacterial strains. The transformation rate of the recombinant bacterium constructed in Embodiment 3 of the present invention was the highest, up to 65.3%.

(25) The above embodiments are only preferred embodiments for a detailed description of the present invention, and the protection scope of the present invention is not limited thereto. Equivalent substitutions or alterations made by those skilled in the art on the basis of the present invention are all within the protection scope of the present invention. The protection scope of the present invention shall be defined by the Claims.

Claims

1. A recombinant *Escherichia coli* for producing glutarate, wherein the recombinant *Escherichia coli* is obtained by recombinantly expressing an aromatic aldehyde synthase (AAS) gene, amine oxidase (Mao) gene and aldehyde dehydrogenase (Glox) gene in a host of *Escherichia coli*.
2. The recombinant *Escherichia coli* according to claim 1, wherein the aromatic aldehyde synthase (AAS) gene, the amine oxidase (Mao) gene and the aldehyde dehydrogenase (Glox) gene are expressed through a double-plasmid expression system, and the aromatic aldehyde synthase gene and the amine oxidase gene are expressed through the same plasmid.
3. The recombinant *Escherichia coli* according to claim 2, wherein the double-plasmid expression system is a combination of two of a pETM6R1 plasmid, pET28a plasmid, a PRSF plasmid, a pCOR plasmid, a pCDF plasmid and a PACYC plasmid.
4. The recombinant *Escherichia coli* according to claim 1, wherein an amino acid sequence of the aromatic aldehyde synthase comprises SEQ ID NO: 1, an amino acid sequence of the amine oxidase (Mao) comprises SEQ ID NO: 2, and an amino acid sequence of the aldehyde dehydrogenase (Glox) comprises SEQ ID NO:3.
5. The recombinant *Escherichia coli* according to claim 1, wherein the host of the *Escherichia coli* is *E. coli* BL21 (DE3), *E. coli* JM109 (DE3) or *E. coli* MG1655 (DE3).
6. A construction method of the recombinant *Escherichia coli* according to claim 1, comprising steps of: S1, ligating the aromatic aldehyde synthase (AAS) gene and the amine oxidase (Mao) gene to a first vector to obtain a first recombinant vector; S2, ligating the aldehyde dehydrogenase (Glox) gene to a second vector to obtain a second recombinant vector; S3, introducing the first recombinant vector and the second recombinant vector into plural initial *Escherichia coli* host cells to produce constructed recombinant strains, and screening the constructed recombinant strains to identify a constructed recombinant strain that is capable of expressing the aromatic aldehyde synthase (AAS) gene, the amine oxidase (Mao) gene and aldehyde dehydrogenase (Glox) gene as the recombinant *Escherichia coli* for producing glutarate.
7. A method for producing glutarate, comprising: providing the recombinant *Escherichia coli* according to claim 1; and reacting L-lysine to produce glutarate in the presence of the recombinant *Escherichia coli* in a catalytic system.
8. The method according to claim 7, wherein a concentration of L-lysine in the catalytic system.
9. The method according to claim 7, wherein the catalytic system further comprises 4-8 nM of NAD.sup.+.
10. The method according to claim 7, wherein the catalytic system has a pH value of 7.5-8.5 and a temperature of 28-32° C., and the reacting is performed while shaking the catalytic system at a rotation speed of 150-250 rpm .